

Novel approach to finding nearest optimal feasible solution to infeasible solid transportation and transshipment problems

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Abstract

This paper explores the formulation and distinctive characteristics of the solid transportation problem, where transportation flow is governed by three sets of constraints: supply, demand, and heterogeneous shipment. A solid transportation problem is deemed infeasible when no feasible solution exists that simultaneously satisfies all imposed constraints. Infeasibility will always occur if the solid problem is unbalanced, but it may also occur otherwise. In such circumstances, we find the closest optimal feasible solution for the unbalanced solid transportation problem, where we minimize both the cost and allocations to imaginary variables needed to balance the problem. This paper presents a multi-stage method of finding such a solution by sequentially formulating two balanced transportation problems and solving for each, where an additional origin, destination, and commodity are added to both problems and the result of the first is used to formulate the second. It also examines the formulation and solution of the multi-index transshipment problem, where transshipment is also allowed between origins and between destinations, as an extension of the unbalanced solid transportation problem. A numerical example along with the network flow diagram is provided to help illustrate the method. To illustrate the algorithm, we present numerical examples of up to order $10 \times 10 \times 3$ for both balanced and unbalanced transportation and transshipment.

Keywords Solid transportation problem, multi-index, transshipment problem, linear programming, unbalanced problem

Numerical Example

For a simple example, we will consider an unbalanced transshipment problem with 1 origin, 2 destinations, and 3 commodities. To illustrate the practical application of the method, we will create a hypothetical situation where an electronic retail chain needs to supply stores with various types of goods. Suppose that two stores located in Los Angeles (LA) and San Francisco (SF) need to restock on different quantities of cell phones, speakers, and televisions ($k = 1, 2, 3$) from a port located in New York (NY). Due to the large distance between the stores in California and the port in New York, it may be economically efficient to make one large shipment from the supply point to the closest store and then transship these resources among the two stores. The supply or demand of each location and the cost of transportation are shown in Table 1 below. Suffix i denotes rows, j denotes columns, and k denotes the diagonal cells within a cell. The bottommost row describes a_{jk} , the demand for each commodity at each destination point; the rightmost column describes b_{ki} , the supply of each commodity at each origin point; and the bottom left corner of each cell denotes e_{ij} , the movement constraint between each row point and each column point. Each internal cell denotes c_{ijk} , the cost of shipping between each row point and each column point for each commodity.

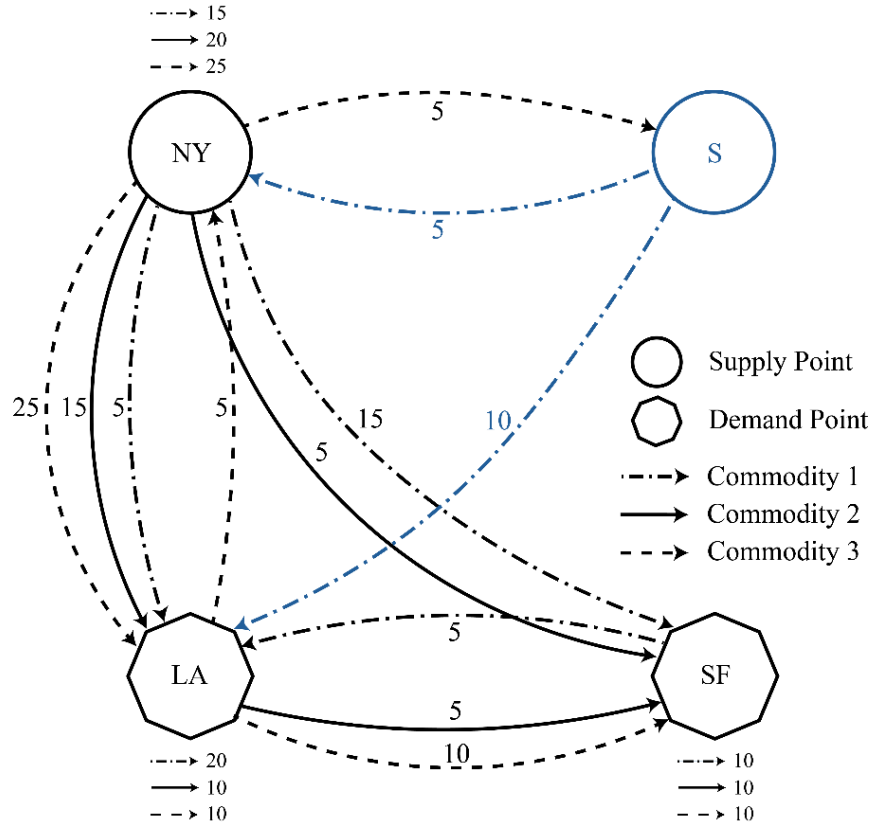


Figure 1: Network diagram of the nearest optimal feasible solution to the practical example

To illustrate the viability of our algorithm on larger problems, we also present a $10 \times 10 \times 3$ transportation problem randomly generated using python. We demonstrate that the algorithm to induce feasibility functions efficiently when the number of supplies and demands are large, and that the produced balanced problem can be efficiently solved using existing Simplex solvers. While the larger problem is too complex to be illustrated graphically, we can create network diagrams of its nearest optimal feasible solution by fixing a specific origin, destination, or commodity. This produces a two-dimensional slice of the problem in the form of a balanced standard transportation problem. To demonstrate the structure of the larger example and show the validity of our solution, we will do this for origin 1, 2, and 3 in Figure 4. All shipments exiting origin 1 are colored blue; shipments exiting origin 2 are colored black; and shipments exiting origin 3 are colored yellow. The number besides each arrow in the diagram is the allocation from the specific origin to the specific destination. Lines above each of the 3 origins (circles) are the supply constraints on each commodity, and lines below each of the 11 destinations (octagons) are movement constraints between the destination and the 3 origins. The demand constraints are not shown.

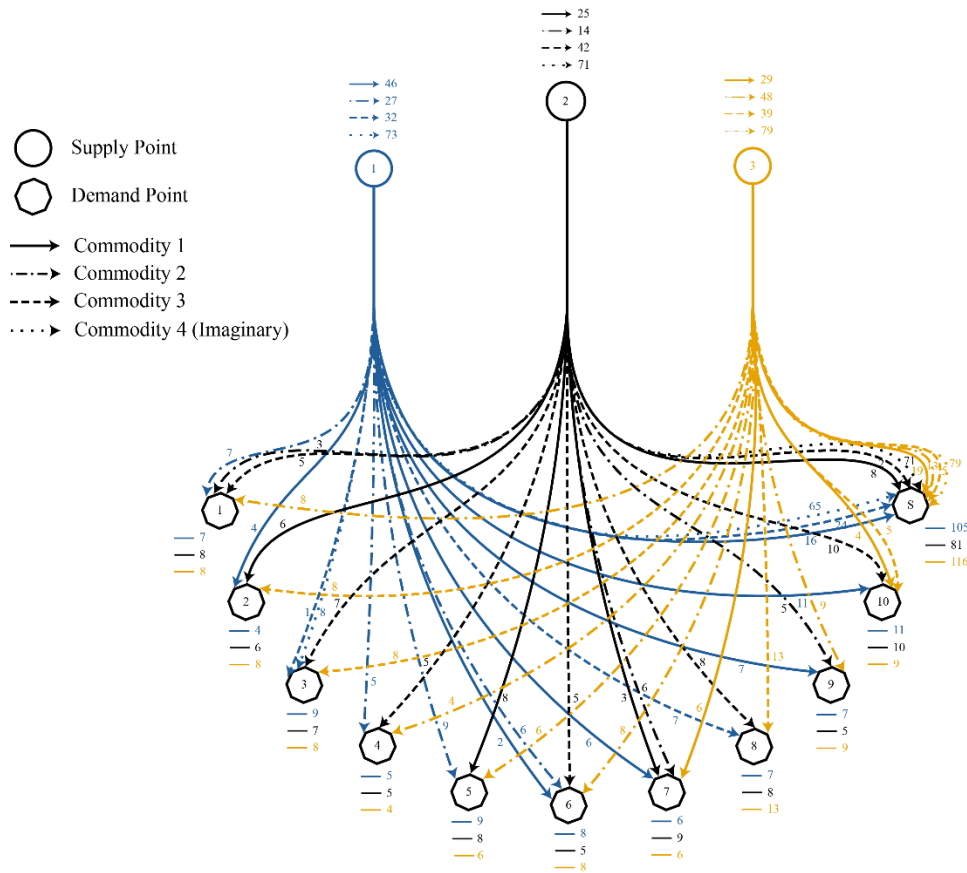


Figure 4: Network Diagram of the optimal basic feasible solution origins 1, 2, and 3 for the large example

Conclusion

In this paper, we discussed infeasibility in solid transportation problems and formulated the nearest optimal feasible solution, in which minimal allocations are added to imaginary variables to produce a feasible problem. The situation of an infeasible solid transportation problem arises when the specific set of supply, demand, and heterogeneous shipment constraints cannot be simultaneously satisfied. As a result, additional supplies, demands, and commodities are created to induce feasibility in the problem. The task of finding the optimal basic feasible solution that most closely satisfies the existing constraints is of interest to managers, as the result outlines the minimal number of modifications to be applied to the system for it to become feasible, as well as how these additional modifications are most optimally satisfied. The specified problem was formulated as a modification of the original problem with constrained flow, at which point it can be transformed into a feasible solid transportation problem with an added supply, demand, and commodity constraint to the original problem and can be solved by the Haley method. The paper also discussed the solid transshipment problem as a variation of the unbalanced solid transportation problem. Furthermore, it provided a method to solve both a balanced solid transshipment problem as well as an unbalanced one using methods similar to the procedure of finding the nearest optimal feasible solution to the solid transportation problem. The transformations and algorithms provided are compact and easy to understand. The findings are useful in areas such as supply chain management, emergency response operations, and online delivery where multiple commodities are transported between a set of locations in which the supply, demand, and transportation capacity are limited and feasibility is not guaranteed. Numerical examples were solved to illustrate the theory, including simple demonstrations and higher order examples that take fractions of a second to solve. For future work, further research can be conducted regarding applying similar transformations on problems where the objective function is to minimize the maximum time of transportation as opposed to overall cost.